

Lecture 3:

Compressive Sensing and Applications

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Outline

Sparsity and representation basis (DCT, Wavelet, Kronecker between DCT and Wavelet)

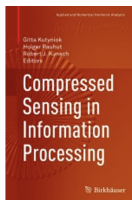
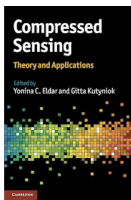
Sensing matrix (Hadamard, Bernoulli, Gaussian, random, blue noise, regular and irregular sphere packing)

Mutual Coherence

Restricted Isometry Property (RIP)

Reconstruction Algorithms:

- * Iterative Hard Thresholding (IHT),
- * Orthogonal Matching Pursuit (OMP),
- * Two-Step Iterative Shrinkage/Thresholding (TwIST),
- * Gradient Projection for Sparse Reconstruction (GPSR)



Notation

- * Matrices and vectors are denoted with **boldface uppercase** and **lowercase letters**, i.e. $\mathbf{A}$, $\mathbf{y}$.
- * The general l_p norm of vector $\boldsymbol{\theta}$ is defined as

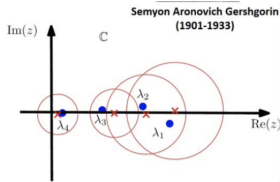
$$\|\boldsymbol{\theta}\|_p = \left(\sum_i |\theta_i|^p \right)^{\frac{1}{p}} \quad (1)$$

$p \geq 1$, where the **Euclidean distance** corresponds to $p = 2$. For $p < 1$ the triangle inequality is not satisfied, and it does not satisfy the definition of a norm.

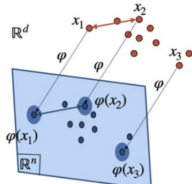
- * A vector $\boldsymbol{\theta}$ is said to be **S-sparse** ($\boldsymbol{\theta} \in \Sigma_S$)
- * $|\boldsymbol{\theta}|_0$ is known as **pseudo-norm**, which gives the number of nonzero elements in $\boldsymbol{\theta}$
- * Optimization problem will be posed in the following form $\hat{\boldsymbol{\theta}} = \operatorname{argmin} f(\boldsymbol{\theta})$ where **argmin** denotes the argument $\boldsymbol{\theta}$ that minimize the expression $f(\boldsymbol{\theta})$.

Timeline Mathematical Technology

Gershgorin Disc Theorem (1931)



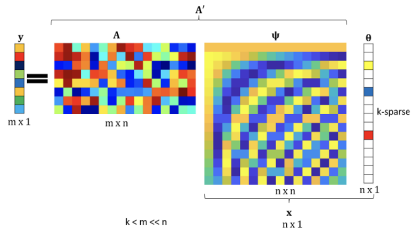
Johnson–Lindenstrauss lemma (1984)



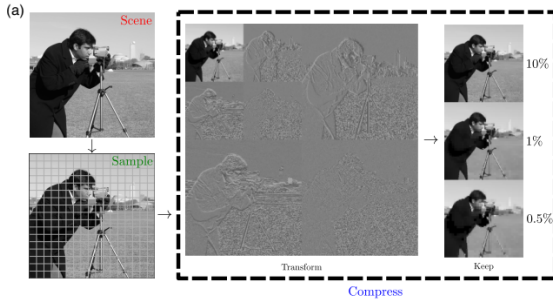
Pseudo-inverse Moore-Penrose (1954)

$$\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y} \rightarrow \mathbf{x} = \mathbf{A}^\dagger \mathbf{y}$$

Compressed sensing (2006)



Conventional sensing against compressive sensing



(b) Conventional Sensing



(c) Compressed Sensing



Sensing types ¹

¹Velat Kilic, Trac D. Tran, and Mark A. Foster, "Compressed sensing in photonics: tutorial," J. Opt. Soc. Am. B 40, 28-52 (2023)

Sampling types

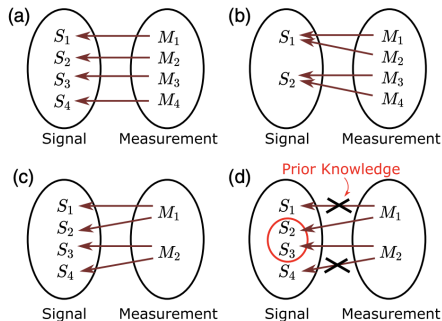


Fig. 2. Measurement M_i to signal S_i mapping for different sampling schemes: (a) critical ("Nyquist") sampling case where mapping is invertible, (b) oversampled case (often employed if measurements are noisy), and (c) undersampled case (compressed sensing). (d) Undersampled mapping can be made invertible by selecting a subset of all possible signals that utilizes prior information about signals (e.g., sparsity).

Single-Pixel Scheme

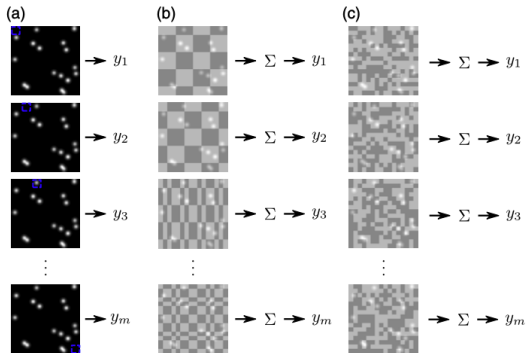
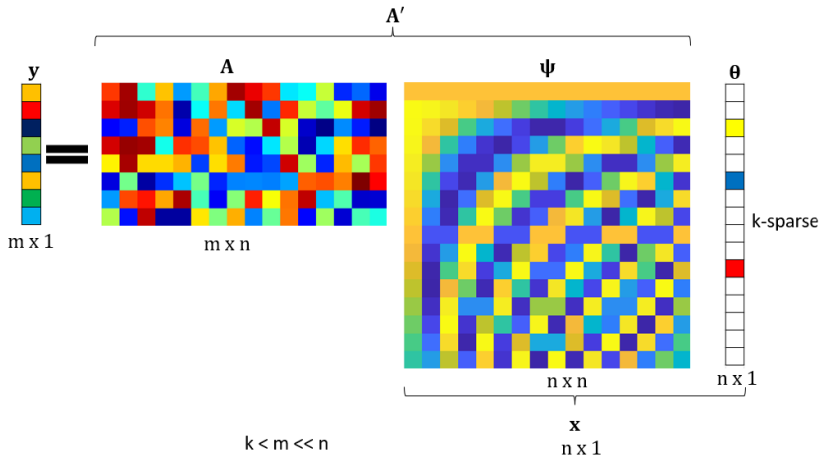


Fig. 3. (a) Raster scanning: a photodetector is scanned through the entire image to acquire independent measurements. (b) Basis scan: independent patterns are projected onto the scene, and the light from the entire scene is integrated by the photodetector. (c) Pseudo-random patterns can be used to optimally sample a scene for compressed sensing.

Compressive Sensing (1)



where y is compressive measurement, A is the sensing matrix, Ψ is the representation basis, and θ is the coefficient vector.

The goal is to recover the signal θ from y

This is an ill-posed problem (Underdetermined system)

Concepts from Linear Algebra

Let $\mathbf{y} \in \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$

$$\mathbf{y} = \mathbf{A}\mathbf{x}$$

Square matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, where $m = n$

Underdetermined system of equation $\mathbf{A} \in \mathbb{R}^{m \times n}$, where $m \ll n$

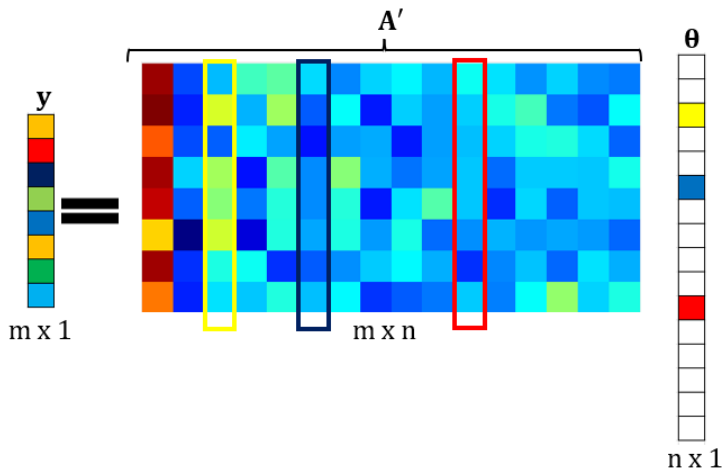
Overdetermined system of equation $\mathbf{A} \in \mathbb{R}^{m \times n}$, where $m \gg n$

Eigenvalues λ and Eigenvectors $\mathbf{v}$

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Null space, denoted $\mathcal{N}(\mathbf{A}) = \{\mathbf{z} | \mathbf{A}\mathbf{z} = \mathbf{0}\}$

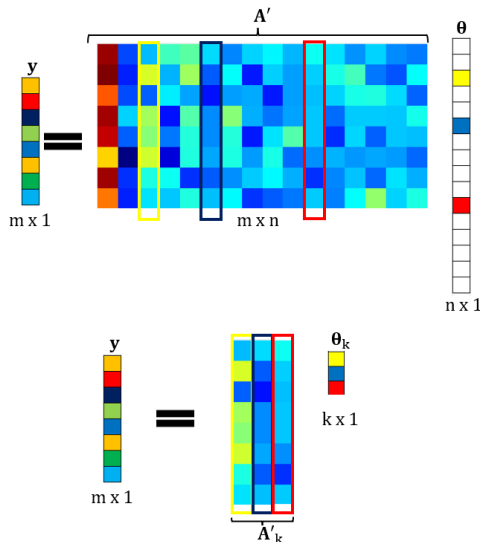
Compressive Sensing (2)



Assuming sparsity $y = A' \theta$

The sparse signal θ select columns of A'

Compressive Sensing (3)



Resulting in an over-determined system $y = A'_k \theta_k$

Norm visualization

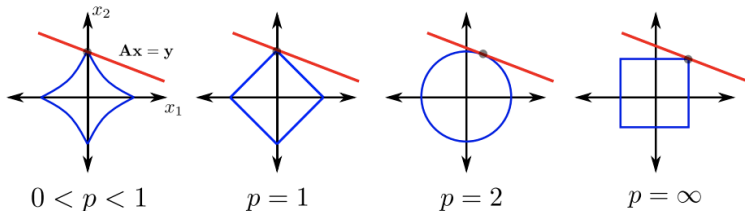


Fig. 5. Red lines show all the solutions satisfying $Ax = y$. Blue lines show the “ l_p balls” that are grown until they intersect with the red line. Gray circles show the solutions from each process where the two lines intersect.

Pseudo-inverse

The case of critical sampling corresponds to $m = N$ and can be solved by $\mathbf{x} = \mathbf{A}^{-1}\mathbf{y}$ if the matrix inverse exists. However, the sensing matrix is not square for the oversampled or the undersampled case and therefore its inverse does not exist. This problem can be circumvented by multiplying both sides of the equation by its transpose $\mathbf{A}^T$,

$$\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{y}. \quad (4)$$

Since $\mathbf{A}^T \mathbf{A}$ is square and can be inverted if $|\mathbf{A}^T \mathbf{A}| \neq 0$, we can solve for the image

$$\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y} \Rightarrow \mathbf{x} = \mathbf{A}^\dagger \mathbf{y}, \quad (5)$$

where the generalized inverse (also known as pseudo-inverse or Moore–Penrose inverse) is defined as $\mathbf{A}^\dagger = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$. The pseudo-inverse is optimal in the sense that it provides the minimal squared error:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{A} \mathbf{x} - \mathbf{y}\|_2^2, \quad (6)$$

Optimization problem

At first glance, the problem $\mathbf{y} = \mathbf{A}\mathbf{x}$ where $\mathbf{A}$ has more columns than rows is underdetermined and, hence, has an infinite number of solutions. Enforcing sparsity on the solution $\mathbf{x}$ is a form of regularization, making the problem better posed and leading to our first optimization problem:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{x}\|_0 \text{ subject to } \mathbf{A}\mathbf{x} = \mathbf{y}, \quad (9)$$

where we attempt to minimize the number of nonzero elements in $\mathbf{x}$. In other words, out of all possible candidates $\mathbf{x}$ that satisfies the constraint set $\{\mathbf{x} \mid \mathbf{A}\mathbf{x} = \mathbf{y}\}$, we propose to select the simplest (sparsest) one, following our original sparsity motivation. This ℓ_0 -minimization problem in Eq. (9) turns out to belong to a

Null Space

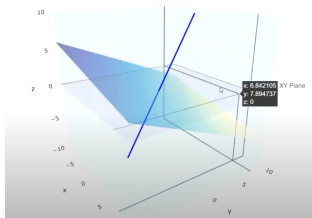
The null space of $\mathbf{A}$, denoted $\mathcal{N}(\mathbf{A}) = \{\mathbf{z} | \mathbf{A}\mathbf{z} = \mathbf{0}\}$ hold the key to the unique recovery.

both $\mathbf{x}$ and $\mathbf{x} + \mathbf{z}$ are possible solutions to $\mathbf{y} = \mathbf{A}\mathbf{x}$

Unique sparse recovery can be successful as long as the null space of $\mathbf{A}$ does not contain any sparse vector.

$\mathbf{A}$ represents all S -sparse signals $\mathbf{x}$ if and only if $\mathcal{N}(\mathbf{A})$ contains no vector in Σ_{2S} , a set of $2S$ -sparse vectors.

Good sensing matrix: Any random subset of $2S$ columns of $\mathbf{A}$ should be linearly independent.



Jhonson-Lindenstrauss lemma (1984)

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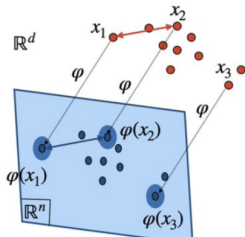
Let $\varepsilon \in (0,1)$, $X = \{x_1, \dots, x_m\} \subset \mathbb{R}^d$, $n \gtrsim (\ln m)/\varepsilon^2$.

Then, there exists a **linear** map $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ such that

$$\forall x, x' \in X, \quad (1 - \varepsilon)\|x - x'\|^2 \leq \|\varphi(x) - \varphi(x')\|^2 \leq (1 + \varepsilon)\|x - x'\|^2$$



W. Johnson



Johnson–Lindenstrauss transform

Constructive proof of existence of φ !

$$\varphi(x) = \frac{1}{\sqrt{n}} Mx$$

with $M \in \mathbb{R}^{n \times d}$, $M_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$



J. Lindenstrauss

EXTENSIONS OF LIPSCHITZ MAPPINGS INTO A HILBERT SPACE

SECTION 8. Theorem 1 and Theorem 2

interest $\mathbf{x}$ might only be approximately S -sparse. The pioneering work by Candès and Tao in [10] introduces the following crucial property called the restricted isometry property (RIP) on matrices $\mathbf{A}$ defined as follows: a matrix $\mathbf{A}$ satisfies RIP of order $2S$ if there exists a bound $0 < \delta_{2S} < 1$ such that

$$(1 - \delta_{2S}) \|\mathbf{x}\|_2^2 \leq \|\mathbf{A}\mathbf{x}\|_2^2 \leq (1 + \delta_{2S}) \|\mathbf{x}\|_2^2 \quad (10)$$

holds $\forall \mathbf{x} \in \Sigma_{2S}$. RIP eventually allows exact sparse recovery of any S -sparse signal via ℓ_1 -minimization as follows:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{x}\|_1 \text{ subject to } \mathbf{A}\mathbf{x} = \mathbf{y}. \quad (11)$$

Figure 5 illustrates at a high level why ℓ_1 -minimization encourages sparse solutions, showing unit circles for different p values (or more accurately l_p balls) in two-dimensional (2D) space

Recovery in presence of noise

In fact, even with approximately S -sparse signal and noisy measurements $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}$ where $\mathbf{w}$ is an additive noise vector, the following generalized optimization also yields robust recovery:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{x}\|_1 \text{ subject to } \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2 \leq \epsilon. \quad (12)$$

We can interpret Eq. (10) as saying that $\mathbf{A}$ approximately preserves the distance between any pair of S -sparse vectors $\{\mathbf{u}, \mathbf{v}\}$ (where we can interpret $\mathbf{x}$ as $\mathbf{u} - \mathbf{v}$). This close relationship between RIP and the Johnson–Lindenstrauss lemma points us to construction of robust matrices $\mathbf{A}$'s as well-understood dimensionality-reduction operators [12]. At closer inspection, one can observe that RIP also has a tight relationship with the aforementioned null-space property of $\mathbf{A}$: any random subset of $2S$ columns of $\mathbf{A}$ yields an approximate isometry, and this degree of isometry δ_{2S} , as expected, will play an important role in the quality of recovery. Therefore, RIP leads to funda-

Is this subspace problem invertible?

Although RIP has nice mathematical properties and guarantees, it is difficult to evaluate in practice since its evaluation is a combinatorial problem. Coherence of a sensing matrix is much easier to calculate and can often elucidate the trade-offs when designing sampling strategies for photonic applications. For successful recovery, knowing the support of $\mathbf{x}$ (i.e., positions of nonzero elements) as shown in Fig. 4 is not enough. We also need to guarantee that the following subspace problem is invertible as well [14]:

$$\mathbf{A}_S^T \mathbf{A}_S \mathbf{x}_S = \mathbf{A}_S^T \mathbf{y}, \quad (14)$$

where $\mathbf{A}_S$ contains columns corresponding to the nonzero elements of $\mathbf{x}$. Therefore, the eigenvalues of $\mathbf{A}_S^T \mathbf{A}_S$ cannot be zero as that would imply $\det(\mathbf{A}_S^T \mathbf{A}_S) = 0$. Intuitively, we can project the signal $\mathbf{x}_S$ onto the eigenvectors of $\mathbf{A}_S^T \mathbf{A}_S$ and observe that an eigenvalue of zero would multiply the signal's corresponding coefficient with zero and information about that component would be irrecoverably lost. Furthermore, a small eigenvalue is not desirable as it might push the signal energy in that mode below the noise floor. Estimating the eigenvalues of $\mathbf{A}_S^T \mathbf{A}_S$ proves invaluable in assessing the reconstruction success.

scheme. The matrix on the left-hand side of the inverse problem measures the overlap of basis functions:

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{a}_1 \rangle & \cdots & \langle \mathbf{a}_1, \mathbf{a}_n \rangle \\ \vdots & \ddots & \vdots \\ \langle \mathbf{a}_n, \mathbf{a}_1 \rangle & \cdots & \langle \mathbf{a}_n, \mathbf{a}_n \rangle \end{bmatrix}, \quad (15)$$

where $\mathbf{a}_i$ are the column vectors of $\mathbf{A}$. The eigenvalues of this matrix will be in a disc centered around the diagonal elements $\lambda_{c,i} = \langle \mathbf{a}_i, \mathbf{a}_i \rangle$ with a radius $R_i = \sum_{j \neq i} |\langle \mathbf{a}_i, \mathbf{a}_j \rangle|$. When solving the subspace problem, $\mathbf{A}_S^T \mathbf{A}_S \mathbf{x} = \mathbf{A}_S^T \mathbf{y}$, successful reconstruction would need eigenvalues of $\mathbf{A}_S^T \mathbf{A}_S$ to be nonzero. In the worst case, all correlated signals would appear in the subspace problem so it is useful to define the *mutual coherence* or simply the *coherence* of the sensing matrix $\mathbf{A}$, $\mu(\mathbf{A})$, as [16]

$$\mu(\mathbf{A}) = \max_{1 \leq i \neq j \leq n} |\langle \mathbf{a}_i, \mathbf{a}_j \rangle|, \quad (16)$$

which is easily computable as the largest off-diagonal element magnitude-wise in $\mathbf{A}^T \mathbf{A}$. Suppose the column vectors are nor-

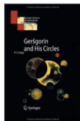
Gershgorin Circle Theorem (1931)

Complex matrix eigenvalues and Gershgorin circle theorem

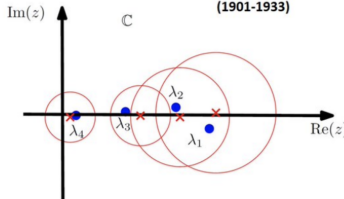
- Complex matrix: $M = [m_{ij}] \in \text{Mat}_{\mathbb{C}}(d, d)$
- Eigenspectrum: $\Lambda(M) := \{\lambda_1(M), \dots, \lambda_d(M)\}$
- Gershgorin circle: $\text{Disk}(c, r) = \{z \in \mathbb{C} : |z - c| \leq r\}$

Theorem (Gershgorin, 1931):

$$\Lambda(M) \subset \bigcup_{i=1}^d \text{Disk} \left(m_{ii}, r_i = \sum_{j \neq i} |m_{ij}| \right)$$



Semyon Aronovich Gershgorin
(1901-1933)



Gershgorin Disc Theorem

The Gershgorin disc theorem provides an easy and intuitive method for bounding the eigenvalues of nearly diagonal matrices. For diagonal matrices, the eigenvalues are the diagonal elements. For matrices that are nearly diagonal, we would expect the eigenvalues to be close to the diagonal values. The Gershgorin disc theorem states that indeed an eigenvalue of a matrix is located in a disc centered around the diagonal element with radius equal to the sum of absolute values of the off-diagonal elements [15] as shown graphically in Fig. 6. It is worth noting that the off-diagonal elements could have been picked from the columns instead of the rows since the eigenvalues of a matrix and its transpose are the same.

The Gershgorin disc theorem might not always give a tight bound (especially if off-diagonal elements are not small), but

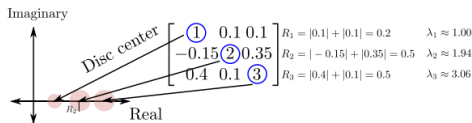


Fig. 6. Applying the Gershgorin disc theorem to estimate eigenvalues of an example 3×3 matrix. Eigenvalues lie on a disc centered around the diagonal elements with radius determined by the sum of absolute values of the off-diagonal elements. Numerically calculated eigenvalues λ_i are indeed located inside the disc.

Iterative Hard thresholding

Algorithm 3. Iterative Hard Thresholding [23]

Input: Sensing matrix $\mathbf{A}$, measurement vector $\mathbf{y}$, sparsity S , error tolerance ϵ

Output: Reconstructed signal $\hat{\mathbf{x}}$

$\hat{\mathbf{x}} \leftarrow \mathbf{0}$

▷ Initialize guess to the zero vector

while $\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{y}\|_2 > \epsilon$ **do**

▷ Termination criteria

$\hat{\mathbf{x}} \leftarrow \sigma_S^H(\hat{\mathbf{x}} + \mathbf{A}^T(\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}))$

▷ Gradient descent followed by hard thresholding

end while

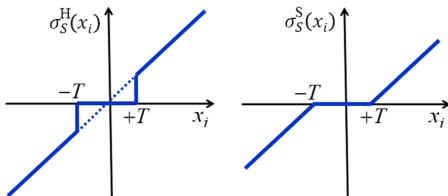


Fig. 7. Thresholding operations are used in iterative convex optimization routines. Hard thresholding (left) simply sets signal elements with absolute value below a certain threshold T to zero while keeping the rest of the values the same. In contrast, soft thresholding (right) scales all the values down by T and then sets elements with small absolute values to zero.

Orthogonal Matching pursuit

Algorithm 1. Orthogonal Matching Pursuit [19]

Input: Sensing matrix $\mathbf{A}$, measurement vector $\mathbf{y}$, sparsity S , error tolerance ϵ

Output: Reconstructed signal $\hat{\mathbf{x}}$

$\hat{\mathbf{x}} \leftarrow \mathbf{0}$

$\mathbf{r}_0 \leftarrow \mathbf{y}$

$\mathcal{S}_0 \leftarrow \emptyset$

while $|\mathcal{S}| < S$ and $\|\mathbf{r}_i\|_2 < \|\mathbf{r}_{i-1}\|_2$ and $\|\mathbf{A}\hat{\mathbf{x}}_i - \mathbf{y}\|_2 > \epsilon$ **do**

$n^* \leftarrow \arg \max_{1 \leq n \leq N} |\langle \mathbf{r}_{i-1}, \mathbf{a}_n \rangle|$

$\mathcal{S}_i \leftarrow \mathcal{S}_{i-1} \cup n^*$

$\hat{\mathbf{x}}_i \leftarrow \arg \min_{\mathbf{x}} \|\mathbf{A}_{\mathcal{S}_i} \mathbf{x} - \mathbf{y}\|_2$

$\mathbf{r}_i \leftarrow \mathbf{y} - \mathbf{A}_{\mathcal{S}_i} \hat{\mathbf{x}}_i$

end while

▷ Initialize guess to the zero vector

▷ Initialize residue to the measurement vector

▷ Initialize support set to the empty set

▷ Termination criteria

▷ Scan for the best basis candidate

▷ Update the support set

▷ The best signal estimate given support set

▷ Update residue

Recent works in compressed sensing and sparse coding have focused on the application of new tools and ideas from deep neural networks. Initial models use a pretrained deep neural network (DNN) $\mathcal{G}_\theta$ (which can be a variational auto-encoder, a generative adversarial network, etc.) as the structural constraint instead of sparsity [35,36]. This generator maps a latent representation or low-dimensional sparse code $\mathbf{z}$ to the signal space: $\mathbf{x} = \mathcal{G}_\theta(\mathbf{z})$. Instead of requiring sparse signals, the deep network $\mathcal{G}_\theta$ implicitly constrains output $\mathbf{x}$ to belong in a low-dimensional manifold via its architecture and the weights adapted from the training data. Effectively, the choice of training data provides the prior knowledge that limits the set of possible solutions shown in Fig. 2(d). In other words, the network can learn the mapping that generates the high-dimensional signal $\mathbf{x}$ in $\mathcal{R}^N$ from a latent low-dimensional sparse code $\mathbf{z}$ in $\mathcal{R}^S$. Often, the sensing matrix can be co-trained with the optimizer here. A minimization process similar to that in compressed sensing is then employed for reconstruction

$$\hat{\mathbf{z}} = \arg \min_{\mathbf{z}} \left\| \mathbf{y} - \mathbf{A} \mathcal{G}_\theta(\mathbf{z}) \right\|_2^2, \quad (26)$$

and the signal-of-interest can be obtained as $\hat{\mathbf{x}} = \mathcal{G}_\theta(\hat{\mathbf{z}})$.

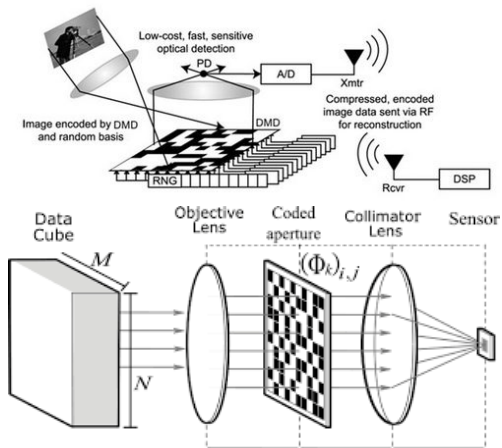
Optical Technologies for Compressed Sensing

- A **Passive Multiplexing:** Uses [static optical devices](#), such as, Coded aperture, Diffusers and Random Microlens Arrays, Scattering, Waveguides and Cavities
- B **Active Multiplexing:** Uses [dynamic optical devices](#), such as, Spatial Light Modulators, Electro-optic Modulators

Applications

- A Single Pixel Camera
- B Microscopy: Fluorescence and Multiphoton Fluorescence
- C On-chip spectrum sensing
- D Hyperspectral imaging
- E Lens-Free Micro-endoscope
- F Optical Coherence Tomography
- G High Speed Imaging
- G Radio Frequency Sensing

Single Pixel Camera



Ultrafast Imaging

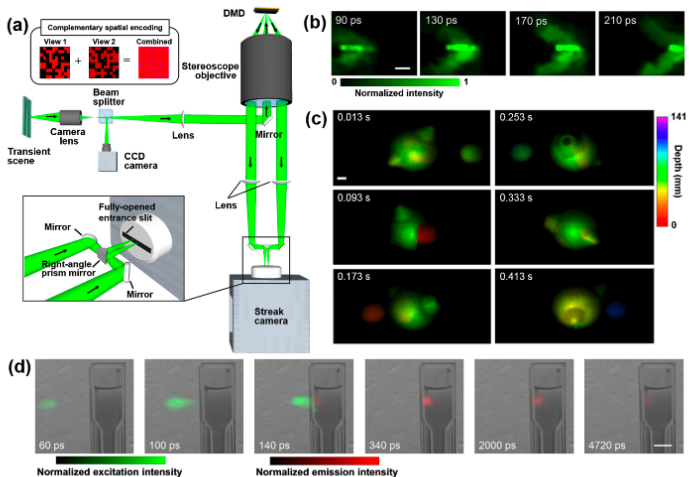


Fig. 22. (a) Compressed ultrafast photography (CUP) system where a DMD is used in conjunction with a streak camera to image high speed events. (b)–(d) Reconstructions for various scenes at high speed. Reprinted from Liang and Wang [106].

Spectral Imaging Applications



Industrial machine vision

- Quality inspection
- Sorting
- Material detection



Medical

- Surgery
- Endoscopy
- Laparoscopy
- Smart pills
- Tissue regeneration monitoring



Agriculture

- Smart farming
- Weed control
- Disease detection



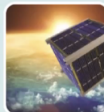
Automotive

- Night vision
- Autonomous cars
- Smart lights



Security

- Intrusion detection
- Camouflage detection
- Shipping monitoring
- Forensic investigation



New Space

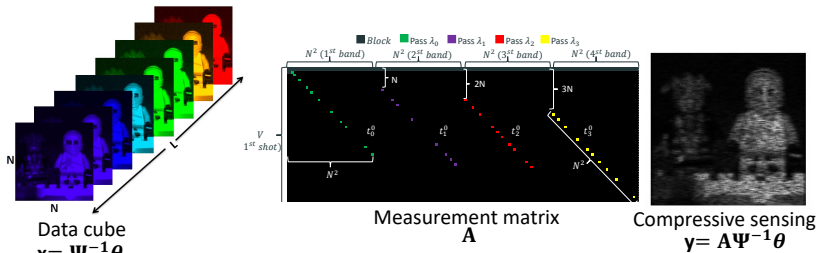
- Global scale image analysis
- Environmental recovery
- Emission monitoring
- Event detection and disaster management



Environment

- Reforestation monitoring
- Oil spill detection
- Plastic detection

Compressive Spectral Imaging

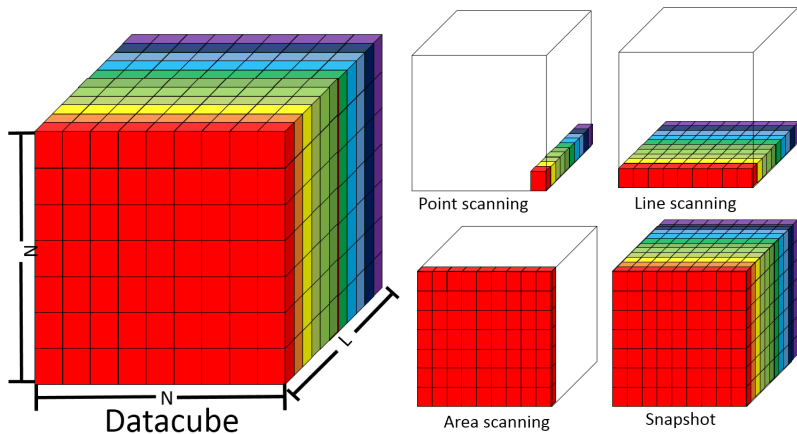


Undetermined system of linear equations

$$\hat{f} = \Psi^{-1}(\operatorname{argmin}_{\theta} \frac{1}{2} \|y - A\Psi\theta\|_2^2 + \tau \|\theta\|_1)$$



Spectral Imaging

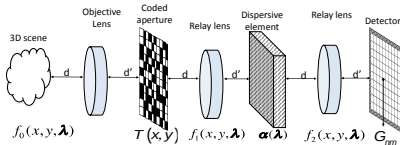


Challenges

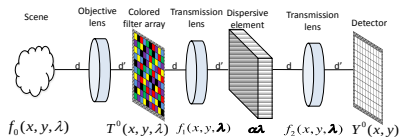
Huge amount of data.

Sequential sampling.

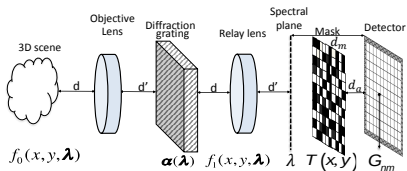
Compressive Spectral Imaging Architectures



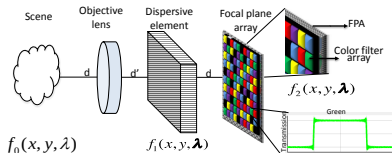
CASSI², 2008, and 2010



C-CASSI³, 2012



SSCSI⁴, 2014



SCCSI⁵, 2015

⁴Wagadarikar, Ashwin, 2008. ⁵Arguello, Henry, 2014. ⁶ Lin, Xing, 2014. ⁷ Correa, Claudia, 2015